

Vikhram Ravi

📍 Chennai, India

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 [LinkedIn](#)

 [Portfolio](#)

SUMMARY

interdisciplinary researcher pursuing a dual M.Tech in Nanobiotechnology (ASNSMM) and MS in Immunobiology (University of Arizona). Bridging the gap between nanotechnology and immunology, my expertise spans bioinformatics and smart nano-formulations. My long-term goal is to leverage these insights to advance personalized medicine, cancer therapeutics, and gene-based therapies. I enjoy troubleshooting multidisciplinary challenges.

EDUCATION

Amrita Vishwa Vidyapeetham, Coimbatore

Amrita School of Nano Sciences and Molecular Medicine, Kochi

Master of Technology, Nanobiotechnology



August 2024 – July 2026

9.48/10

University of Arizona, USA

Department of Immunobiology, College of Medicine, Tucson

Master of Science, Immunobiology



December 2024 – May 2026

4/4

Saveetha Institute of Medical and Technical Sciences, Chennai

Saveetha School of Engineering, Chennai

Bachelor of Technology, Biotechnology



July 2020 - June 2024

9/10

National Institute of Open Schooling, India

Open Schooled, Chennai

Senior Secondary, Physics, Chemistry, Biology, Computer Science



June 2018 - May 2020

78.6 %

Central Board of Secondary Education, India

Vikhe Patil Memorial School, Pune

Secondary, Class X



June 2017 - May 2018

6.72/10

EXPERIENCE

NextGen Biology Newsletter, Visakhapatnam

Volunteer

- Dynamic Support, Content Editing and Proofreading.



February 2026 - Present

Remote

Sri Projects, Chennai

Intern

- Biogas yields of various substrates, VBA Macros involving various conditions / real time scenarios from field data, Biogas Output Mass Balance Calculations for Biogas Plants.
- Basics of AFT Fathom for fluid dynamics analysis in Biogas Plants, Stress Analysis for Bio-related plants using CEASAR II.



February 2024 - July 2024

On-Site

NyBerMan Bioinformatics, France

Intern

- Analyzing the data from COSMIC database for Breast cancer mutations.



February 2024 - July 2024

Remote

- Molecular Docking to identify a suitable drug which is effective against specific mutations.

TECHNICAL SKILLS

Characterization	SEM, DLS/ZETA, XRD, H-NMR, FTIR
Bioinformatics	Molecular Docking, PyMOL, DSV, Binding Site Prediction, BLAST, MD Simulation, Linux
Polymer Chemistry	Linear Polymer Synthesis, Carothers Equation
Gene Delivery	Polyplex systems, Core-Shell systems
Molecular Biology	DNA/RNA Extraction, Plasmid Isolation, TLC, SDS PAGE, ELISA, ROS/Toxicity Assays
Cell Culture	L929, RAW 264.7, HEK 293, PBMC Isolation, Transfection Experiments, in vitro Assays
Statistical / Data Analysis	Origin Lab, SPSS, GraphPad PRISM, Tableau
Industrial Biotechnology	Biogas Mass Balance, PFD, AFT Arrow, CAESAR II Stress Analysis
Documentation	MS Office Suite, LaTeX, Markdown, VBA Macros

SOFT SKILLS

Conflict Resolution	Flexible & Adaptive
Time Management	Listening
Process Optimization & Automation	

PROJECTS

Master's Thesis	June 2025 - Ongoing
A Novel Non-Viral Gene Vector: PBAE based Core-Shell Nanoparticle System for Gene Delivery	
• Synthesis of PBAE, Troubleshooting approaches and Optimization • Core-shell system design, Cell Culture, GFP-Transfection studies (in progress)	
Bachelor's Thesis	May 2024
Examining the Therapeutic effects of Camphene Conjugated with Zinc Oxide Nanoparticles on Zebrafish Larvae	
• Developmental Toxicity Studies in Zebrafish in-vivo model. • Neurotoxicity analysis by using locomotory parameter. • Effects on ROS Enhancement Levels using GPx and GST parameters. • Lipid Peroxidation level analysis using DCFH-DA staining fluorescent Assay	
Obsidian- Personal Knowledge Database	May 2024
• I've created a personal knowledge management system for my coursework using Obsidian. The topics have storyboard layout to help connect related notes with a mind map approach. • The list of topics is categorized and available in my personal portfolio under projects section. (Note: The HTML canvas export has some bugs; raw files are available upon request.)	
GenAI Project (Hobby/Personal)	May 2024
Caraoke , Offline Karaoke app for syncing Lyrics with song	
• Source code was developed using AI with prompts. • APK was built using Android Studio.	
Mini Projects	
Ammo Mist , Ammonia sensor for Aquariums	October 2023 - March 2024
• Developed a module with IoT support for detecting dissolved ammonia in Aquariums • The setup is different from the conventional electrode-based ammonia detection. • Received Institutional Patent for the invention.	
Clin Mist , Natural Surface Disinfectant	July 2022 - December 2022

- A novel surface disinfectant utilizing the antimicrobial properties of terpinen-4-ol extracted from Green tea extract.
- The product received institutional patent

COURSES & CERTIFICATIONS

Fundamentals in Biotechnology and Genetics

Issuer: Alison (NPTEL) 



June 2024

- DNA tools, Basics of Heredity and Genetic terms, Mendelian Genetics

Google Business Intelligence

Issuer: Google (Coursera) 



April 2024

- Obtained the Professional Certificate by completing the three courses, Foundations of Business Intelligence, The Path to Insights: Data Models and Pipelines, Decisions, Decisions: Dashboards and Reports

Drug Discovery

Issuer: UC San Diego (Coursera) 



March 2024

- The phases in the development of drug, parameters involved in bringing a drug from the laboratory to the market.

Introduction to the Biology of Cancer

Issuer: Johns Hopkins University (Coursera) 



March 2024

- Obtained the certificate with Honors by completing the optional course on Introduction to Liver Cancer

Drug Delivery: Principles and Engineering

Issuer: IISc Bangalore (NPTEL) 



October 2022

- Obtained the certificate with Honors by completing the optional course on Introduction to Liver Cancer

CONFERENCES & WORKSHOPS

Virtual Winter School on Computational Chemistry, GROMACS

Jan 2026

- Engaged with training modules led by the GROMACS Team and the Virtual Winter School for molecular docking simulations

8th Indian Myeloma Congress, Annual Conference of IMAGE, Amrita Institute of Medical Sciences, Kochi

Jan 2026

- Attended the workshop on Flow Cytometry for MRD, Demonstration of Gating strategies and viewpoint on MRD clinical decisions based on NGF. Expert viewpoint on Flow Cytometry vs NGS.

Workshop on “Electrospinning for Nanofibre production”, Sathyabama, Chennai

May 2024

14th International Conference on Science & Innovative Engineering, Chennai

April 2024

- Paper Presentation titled, "Evaluating the Pharmacological Profile of Novel Drug Camphene conjugated with Zinc Oxide Nanoparticles using Zebrafish Larvae model".

Workshop on “Biomass Fermentation and Distillation for Bioethanol Production”, Sathyabama, Chennai

May 2024

Workshop on “Biomimetic Scaffolds Fabrication for Tissue Engineering using Novel Electrospinning Strategies”,

Binghamton University

June 2021

PUBLICATIONS

Springer – BioNanoScience

September 2024

Investigating the Protective Effect of Camphene-Conjugated Zinc Oxide Nanoparticles Against *Pseudomonas aeruginosa* Infection in *Danio rerio*

DOI : [10.1007/s12668-024-01561-3](https://doi.org/10.1007/s12668-024-01561-3)

AWARDS

GATE Biotechnology 2026 - Qualified, IIT Guwahati

March 2026

Institute Scholarship, ASNSMM, Amrita Vishwa Vidyapeetham

July 2024 – Present

- 2nd Rank Holder of the Nanobiotechnology Department.

Best Outgoing Student, SSE, SIMATS University

November 2024

- Best Outgoing Student of 2024 Batch
- First mark in Plant Biotechnology and Molecular Biology

CHES:

VALOR represented team Zelos, ASNSMM, 1st Place

March 2025

SIMAM represented team Samudra, SIMATS, 1st Place

May 2022

CBSE Nationals represented Jawahar Vidyalaya, 2nd Runner Up

November 2014

CBSE Zonals represented Jawahar Vidyalaya, 1st Runner Up

October 2014

REFeree

Dr. Dhanya Narayanan (*Assistant Professor, ASNSMM, Kochi*)

dhanyan@aims.amrita.edu

Dr. Nafees Ahmad (*Professor, Immunobiology, UoA, Tucson*)

nafees@arizona.edu

Dr. I. Praveen Kumar (*Assistant Professor, SIMATS, Chennai*)

praveenkumarissac@gmail.com